TEST REPORT - WOOJINFLAME (SCREW BARREL, SCM415)

Header

Customer No.	1083
Product Line	4 - Milling
Industry	F5 - Other Machinery
Test Date (DD/MM/YYYY)	15/03/2026
Test Reason	1 - Productivity
Stage	3 - Completed
Test Result	1 - Positive
Branch	SEOUL

Machine Information

Type	8 - Multi Tasking Mill / Turn
Brand & Model	MAZAK INTEGREXi-400S
Adaptation	HSK A 63
Max. RPM	6,000

Cutting Condition

		Actual	Proposal 1	Proposal 2	Proposal 3	Proposal 4	Proposal 5
Manufacturer		1 - Iscar	1 - Iscar	1 - Iscar			
Tool		FFQ4 D16/.63-03-MMT10-06	FFX4 ED16/.63-2-MMT10-04	MM S-A-L150-C16-T10-W-H			
Diameter (mm)		16	16	16			
No. of Flutes		3	2	4			
Insert		FFQ4 SOMW 060213T	FFX4 XNMU 040310RM-T	MM EFF160T5R3.25-4T10			
Carbide Grade		IC808	IC808	IC903 + TiAlSiN 코팅			
Speed	Cutting Speed (m/min)	90	50	50			
	Spindle Speed (rpm)	**1791**	**1000**	**1000**			
Feed	Feed per Tooth (mm/t)	0.04	0.1	0.06			
	Table Feed (mm/min)	**215.0**	**200.0**	**250.0**			
Width of Cut (mm)		10	10	10			
Depth of Cut (mm)		0.7	0.5	0.5			
MRR (Q, cm3/min)		**1.5**	**1.0**	**1.3**			
Coolant		2 - External	2 - External	2 - External			
Cutting Time							
Tool Life (pcs)		0.7EA	0.7EA	2.5EA			
Spindle Load (%)							
Remark							

Material & Fixture

Part Name	SCREW BARREL
Raw Material	14 - Pre-Machined
Group	9 - Low alloy steel and cast steel (less than 5% all element)
Std. Code	8 - JIS
Material	SCM 415
Hardness	62
Unit	HRC
Rigidity	2 - Semi Stable

Test Result

1. SCREW BARREL 신규개발품 포켓팅 공정 테스트 진행 / 홀더 MM S-A-L150-C16-T10-W-H 텅스텐 사양
2. 소재 SCM415 + 열처리 (HRc 62) 소재 및 클램핑 불안정한 조건으로 FFQ4, FFX4 사양은 소재 1EA 가공중 치핑 발생
3. MM EFF160T5R3.25-4T10 903 + TiAlSiN 코팅 기준 공구수명 2.5EA 확인되어 최종 양산 적용 완료
4. 헤드 22EA 발주

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